

UTD- Fall 2025

CS 4347 Database Systems

Database Project Description

Introduction:

You will have the chance to investigate and gain practical experience with database design and programming through a project. To provide common functionalities like record inserting, updating, deleting, browsing, and searching and filtering, the project will involve designing a database, complete with all supporting evidence of its sound design, as well as front-end and back-end components.

The complexity, difficulty, scale, and theme of a good team project can vary significantly. The sole prerequisite we established for this project is that it must be related to the course material we have been learning. MySQL should be used to generate and query your database, and PHP or any other tool that you are familiar with should be used to construct an interface that users and administrators can use to access and/or modify the database.

Team Structure:

The project is a team project for students to learn how to become effective team players in a software project. Each team may have **four to six** students, and one student is dedicated as the team leader. Each team member must contribute to all project deliverables.

Important Note: To be successful in doing your database project for this course, you must initially familiarize yourself with installing and learning how to connect HTML to a database with MySQL using PHP. There are so many available online tutorials and videos that may guide you in this process.

Project Phases and Deliverables Items

Phase 0 (5%): (Due on Sunday 09/07/2025) Teams Formation:

By this date (Sunday, September 07, 2025), you should have formed your teams. Each team must submit the team's name and the members of the team in which you provide each member's name and his/her student ID.

Phase 1 (15%): (Due on Sunday 09/21/2025)

You must provide a report that includes an overview of your project idea. As we progress through the course, this summary may shift a little, but the overarching theme should remain the same.

The proposal should have two (2) or more pages that outline the problem you intend to address, how you intend to solve it, and what you intend to "deliver" for the finished project. The sections listed below should be included in your report:

1. **Project details:** Name of your project, your team, and team members. Please include Members' IDs.
2. **Problem Statement:** In what ways will your suggested database system solve the issue? Why would you need a database as opposed to an Excel document?
3. **Target user:** Who will access the database? Who is going to manage the database? Giving a real-world example is suggested.

4. **List of Relations:** Choose at least four relations (tables) that you must keep up with for this database. Give schemas for every one of these relationships.
5. **Web interface:** Describe the web interfaces you'll be designing in preliminary designs. Web interfaces often consist of a "form," where a user can enter data and based on that data, the tables are updated, or query results are generated. Your project's user interface looks like this.
6. **Data:** How will you populate your relations? You can get the data from external sources, or you can create your own data.

How to submit:

Save your report as Phase1Report.pdf and upload it via eLearning. Only the team leader needs to submit this file via eLearning.

Phase 2 (20%): (Due on Sunday 10/12/2025).

Our next phase is designing the database. All you need to do for this part is to provide an ER (or EER) model for your database and then convert it into relations (tables).

Task A: Draw an ER (or EER) diagram.

The ER (or EER) diagram should accurately represent the proposal you have provided. If new entities are needed, you should add (or edit) them after starting with the same relations that you described in your proposal (in Phase I). The entities, their characteristics, and their connections (relationships) must all be identified.

For each entity (for each relation), provide total participation, and (min, max) constraint.

There are many utilities on the web for drawing these ER (or EER) diagrams. For example <https://creately.com/> and www.draw.io.

Task B: Convert your ER diagram into relations.

Once the ER (or EER) diagram is ready, you need to convert the ER (or EER) model into a relational database schema.

Your report should have the following components:

1. A list of relations and their attributes.
 - Indicate the primary key and foreign keys (if any) for each relation.
 - Write a detailed description of each attribute (for each table), its purpose, and data type.
 - Indicate each attribute's default value (if any) or if the attribute can be set to 'null'.
 - Explain the actions taken on any foreign key when the primary key (that the foreign key refers to) is deleted (For example no action, delete cascade, set null, or set default).
2. A mapping from the ER (or EER) diagram to the Relations.

How to submit:

Save your report as Phase2Report.pdf and upload it via eLearning. Only the team leader needs to submit this file.

Phase 3 (30%): (Due on Sunday 11/16/2025)

In the first part of this phase, you will provide a list of all the relations you are using, along with any functional dependencies they have. In the event that the relations are not in Third Normal Form normal form (3NF), you will then convert them into this form.

In the second part of this phase, you will mostly work with HTML, PHP, and SQL. For the first time, you will convert your database's logical design into a physical design. You're going to provide "real" interfaces for dealing with the database at this last stage.

Task A: Third Normal Form Normalization

You must ensure that all of your relations are in Third Normal Form for this assignment. If any table is not in 3NF, then normalize it into 3NF. All of the resulting relations should be in 3NF after the decomposition.

Make a file named TaskA.pdf that details the conversion of the basic schema to the finished schema with all relations expressed in 3NF. All the functional dependencies you began with should be included in this file as well. It's possible that your original schema is already represented in 3NF; in that case, all you'd need to do is to show the functional dependencies and that the relationships are already represented in 3NF.

Task B: Create forms using HTML.

You have offered your web interface as graphics in phase I. You will deliver these as HTML files this time. Give a list of HTML files for the interface. These interfaces are primarily employed to obtain user input. Both database connectivity and PHP code are not necessary for this task. We are solely considering the interface. If necessary, you may also include JavaScript files for input validation and CSS files for styling.

- For this stage, CSS and JavaScript are not required.
- Provide HTML files that will be filled with the results of queries. There is no need to connect to the databases to present results for this task; only note that you will do so later.

Create a file called taskB.pdf and add all the HTML, CSS, and JavaScript (if any) files to it.

Task C: Creating and loading relations.

The relations (tables) needed for your project were created in Phase 2. In this stage, you will establish the actual relationships. To construct all the tables in your database, create the file create.sql. The data files for these relations should be tab- or comma-delimited. You have the option of creating the tab- or comma-separated files yourself (using dummy values) or providing the sources. To load data in bulk, create a load.sql file.

Create a file named taskC.pdf and add create.sql, load.sql, and all of the.dat files (or.csv files, or data files in any other format) to it.

How to submit

A new folder called Phase3 Report should be created. Into this folder, upload the files taskA.pdf, taskB.pdf, and taskC.pdf.

Submit this folder via eLearning.

Only the team leader needs to submit this folder.

Phase 4 (20%): Due on Monday (12/01/2025) thru Thursday (12/04/2025)

For this phase, just demonstrate this project to the TA during the final demo period.

The last phase of your project requires you to produce a fully functional website that can communicate with the underlying database system and carry out necessary CRUD (create, read, update, delete) actions.

A few HTML files represented the forms that were delivered in phase. The database and these forms might not have communicated to one another. For these HTML files to function, the required corrections must be made. It should be possible to use all the reports and forms that display the results of queries. When necessary, your website must validate user input and flag any mistakes.

Phase 5 -Final Complete Project Report (10%) (Due on Friday 12/04, 2025)

Submit a well-written and complete report that includes the work that you have done in all the previous phases (one through four). The contents of the Final Project Report should include, but not be limited to the following sections:

Only the team leader needs to submit this final file.

1. Cover page

Provide the title of the course, the title of the project, the name of the instructor, the names of the team members, and the date.

2. Table of contents

Show the contents of the report and their corresponding page number. **3.**

Introduction

Provide a brief description of the project and the section organization of this report.

4. System Requirements

Give the context diagram (system architecture diagram) of the database system. List the interface requirements of the system (or each subsystem).

List the functional and non-functional requirements of the database system.

5. Conceptual Design of the Database

The complete Entity-Relationship (ER) model of your database.

The data dictionary and business rules (i.e., constraints) of your ER model.

6. Logical Database Schema

Give the schema of the database which is restructured and translated from the ER diagram presented in the section "Conceptual Design of the Database". Show the schema with appropriate referential constraints. Give the SQL statements used to construct the schema. List the expected database operations and estimated data volumes.

7. Functional Dependencies and Database Normalization

Identify and analyze the functional dependencies for each relation presented in the section "Logical Database Schema".

Show the normalization process and normalized tables for each relation to 3NF (if applicable). Give the SQL statements for constructing the normalized table (if applicable). **8. The Database System** Give a brief description of how to install and invoke your system.

Provide the “screen dumps” showing how to use your system step by step.

9. Suggestions on Database Tuning (optional)

Give suggestions on tuning your database in terms of index structures, database design, and/or queries.

10. User application interface: describe how you build the system user interface and how users use your system.

11. Conclusions and Future Work

Give a conclusion or your feedback about this project. Provide a brief description of possible future work.

12. References

List the references or books used for this project.

13. Appendix

An appendix gives the zip file containing the work products (including demo slides, final report, SQL scripts, and source code of the project).

The zip file must have the following directories for all the teams:

/doc (contain all documents and presentation slides)

/project (contain all source code, test code, data, web pages, SQL scripts, library, and executable files)

a README file (describing how to install and use your program)